

Unnamed_Unit

Unit 4 - Code Quest

PDF Documents

PDF Document 1: 1738062179718-Chapter 4 - unit 4 - Code Quest.pdf

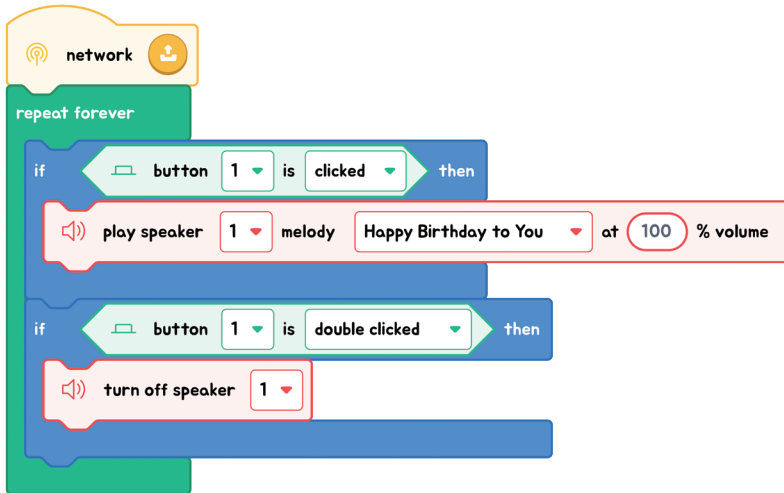
This PDF document is included in the unit materials.



UNIT 4 CODE QUEST

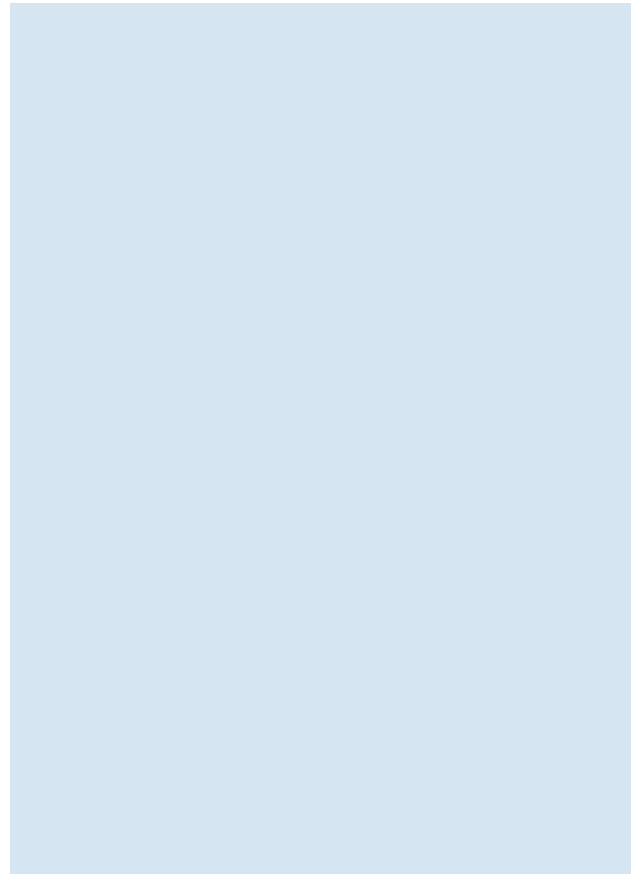
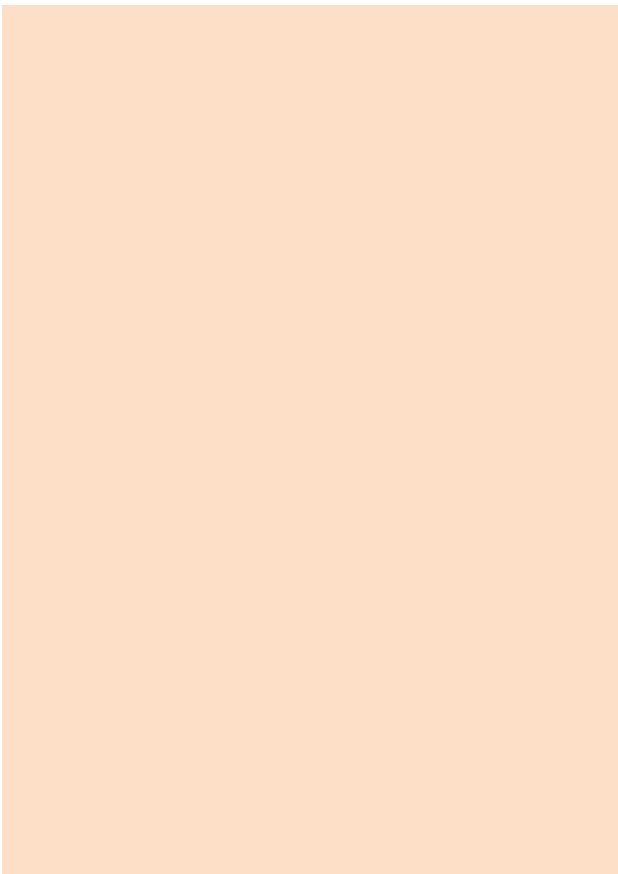
Applying Knowledge (code & debugging)

CODE



DEBUGGING

- 1 What if we don't have a music off function??
- 2 What if we changed Button's option to toggle?



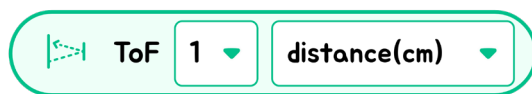
Missions

Mission1

1 Let's change the volume of music. Find a block that fits in the blank space



Example.



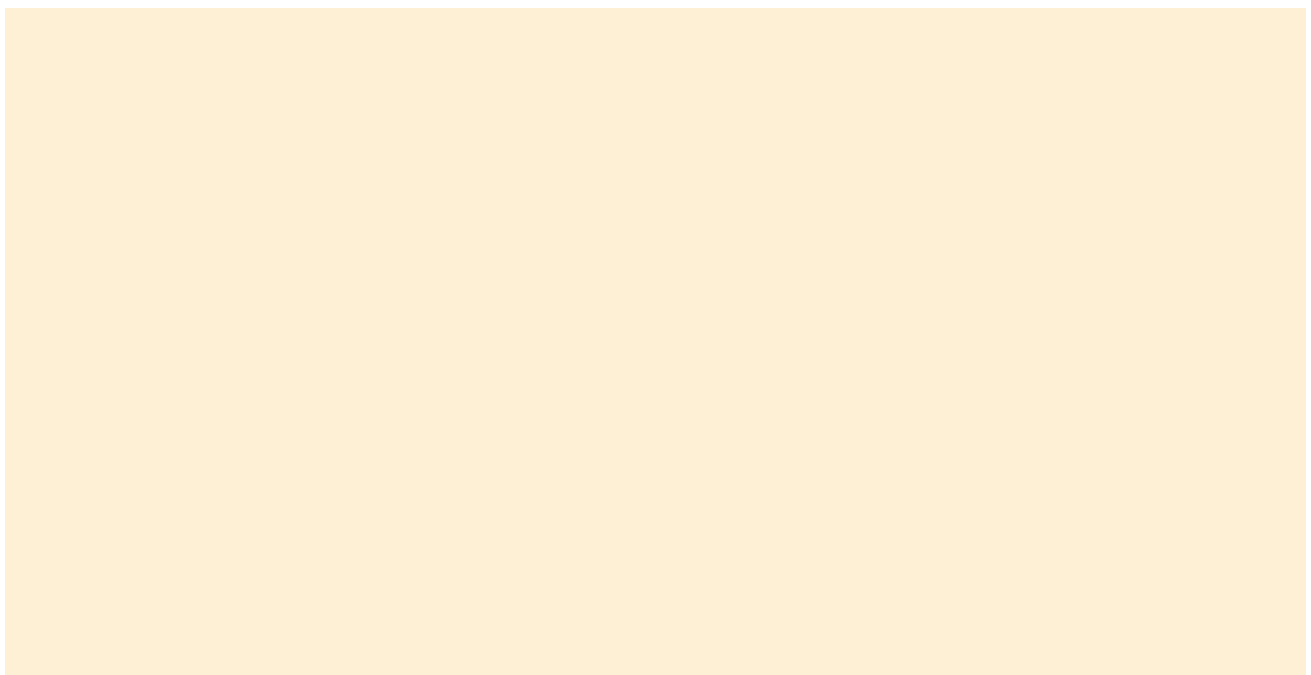
Dial - turn

Joystick - x position

IMU - y angle

Environment - illuminance

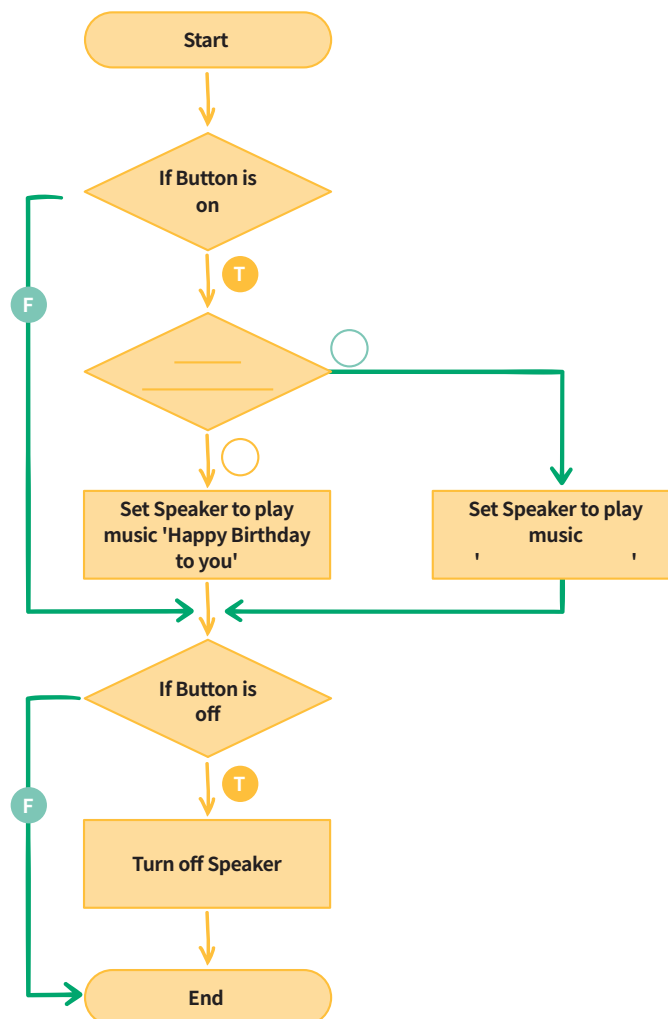
2 Use the block to make the code to change the volume.



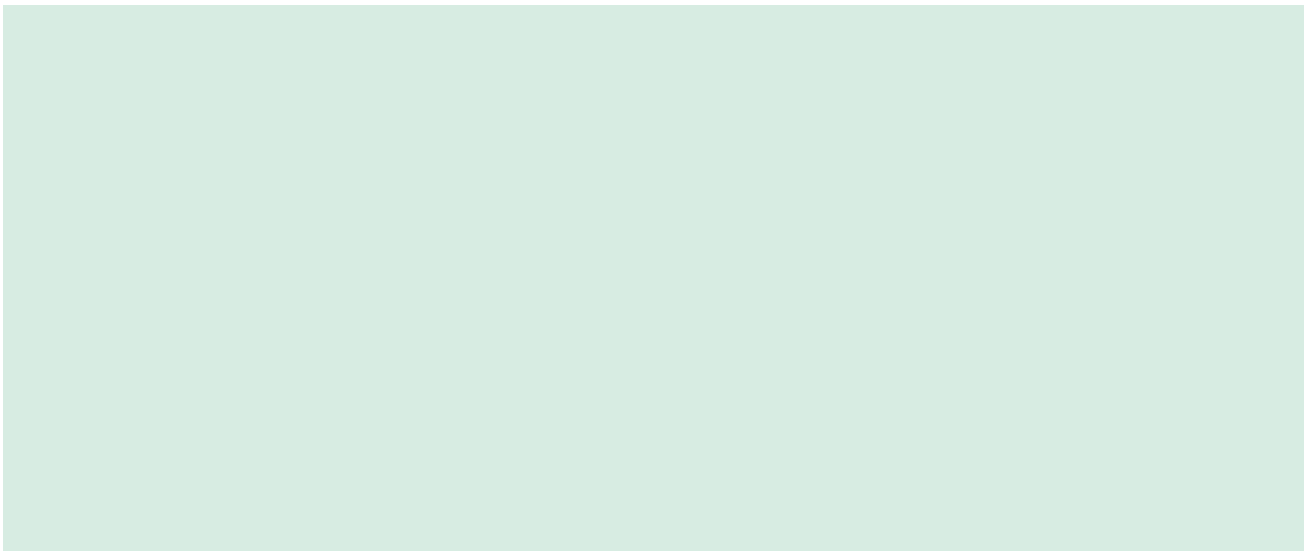


Misson2

- 1 Let's play different song if Dial's turn is over than 50% (use nested-if statement)



- 2 Based on flowchart make the code



KNOWLEDGE CHECK



1 What is music?

- ☐ A A type of food
- ☐ B A form of art that uses sounds
- ☐ C A type of dance
- ☐ D A form of exercise

CORRECT ANSWER

☐ B

2 What is a scale in music?

- ☐ A A measuring tool
- ☐ B A type of fish
- ☐ C A set of musical notes arranged in order
- ☐ D A type of musical instrument

CORRECT ANSWER

☐ C

3 How many notes are there in a standard musical scale?

- ☐ A 5
- ☐ B 7
- ☐ C 10
- ☐ D 12

CORRECT ANSWER

☐ B

5 What is the purpose of a nested-if in programming?

- ☐ A To check 2 or more conditions are met
- ☐ B To define a function in the program
- ☐ C To execute a block of code only if a certain condition is true
- ☐ D To declare variables and assign values to them

CORRECT ANSWER

☐ A

6 Which keyword can be replaced nested-if in most programming languages?

- ☐ A for
- ☐ B while
- ☐ C and/or
- ☐ D else

CORRECT ANSWER

☐ C

7 Which of the following is NOT a function of Button?

- ☐ A Clicked
- ☐ B Double clicked
- ☐ C Turn
- ☐ D Pressed

CORRECT ANSWER

☐ C

8 Which shape in a flowchart is typically used to represent an action?

- ☐ A Rectangle
- ☐ B Diamond
- ☐ C Circle
- ☐ D Parallelogram

CORRECT ANSWER

☐ A



Critical Thinking Challenge

1 What is your favorite song, and in what situations do you enjoy listening to it?

2 Why do we need nested-if statement?